THE POLARIZED STACK IS A STACK IN GROUPOIDS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Definition stacks-definition-topology-inherited (canonical tag 06NV). cited at statement line 4087. Inherited topology on the base category.
- Definition stacks-definition-stack-in-groupoids (canonical tag 02ZI). cited at proof line 4095. The three stack conditions.
- Lemma lemma-polarized-fibred-in-groupoids (canonical tag 0D3Z). cited at proof lines 4096-4097. Condition (1) already known.
- Proposition spaces-descent-proposition-fpqc-descent-quasi-coherent (canonical tag 04W8). cited at proof lines 4127-4128 and 4151-4152. Descent of quasi-coherent modules gives unique gluings.
- Lemma spaces-descent-lemma-descending-property-ample (canonical tag 0D3C). cited at proof lines 4160-4161. Ampleness descends.
- Lemma lemma-spaces-stack (canonical tag 0D1G). cited at proof lines 4164-4165. The base is a stack in groupoids over Sch_fppf.
- Lemma stacks-lemma-stack-over-stack (canonical tag 09WX). cited at proof lines 4166-4167. Formal transfer to Sch_fppf.

Source reference. Original definition 06NV: sources/stacks-chow/tags.txt, source lines 8549–8549. Cited at statement line 4087: the topology on Spaces'_{fp, flat, proper} is the inherited one; load-bearing for the first assertion.

Source reference. Original definition 02ZI: sources/stacks-chow/tags.txt, source lines 3913–3913. Cited at proof line 4095: conditions (1), (2), (3) being checked; load-bearing.

Source reference. Original lemma 0D3Z: sources/stacks-chow/tags.txt, source lines 16434–16434. Cited at proof lines 4096-4097: condition (1) already verified; load-bearing.

Source reference. Original proposition 04W8: sources/stacks-chow/tags.txt, source lines 6351–6351. Cited at proof lines 4127-4128 (descent for morphisms) and 4151-4152 (descent for objects): unique gluing of isomorphisms and of the invertible module; load-bearing twice.

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Source reference. Original lemma 0D3C: sources/stacks-chow/tags.txt, source lines 16412–16412. Cited at proof lines 4160-4161: the descended L is ample; load-bearing.

Source reference. Original lemma 0D1G: sources/stacks-chow/tags.txt, source lines 16346–16346. Cited at proof lines 4164-4165: base stack is a stack in groupoids; load-bearing.

Source reference. Original lemma 09WX: sources/stacks-chow/tags.txt, source lines 12510–12510. Cited at proof lines 4166-4167: formal transfer of the stack property to Sch_fppf; load-bearing.

Lemma 0.2. *The category $\mathcal{P}olarized$ is a stack in groupoids over $\mathcal{S}paces'_{fp,flat,proper}$ (endowed with the inherited topology, see Stacks, Definition 06NV). The category $\mathcal{P}olarized$ is a stack in groupoids over Sch_{fppf}.*

Proof. We prove *Polarized* is a stack in groupoids over $\mathcal{S}paces'_{fp,flat,proper}$ by checking conditions (1), (2), and (3) of Stacks, Definition 02ZI. We have already seen (1) in Lemma 0D3Z.

A covering of $\mathcal{S}paces'_{fp,flat,proper}$ comes about in the following manner: Let $X \to S$ be an object of $\mathcal{S}paces'_{fp,flat,proper}$. Suppose that $\{S_i \to S\}_{i\in I}$ is a covering of Sch_{fppf}. Set $X_i = S_i \times_S X$. Then $\{(X_i \to S_i) \to (X \to S)\}_{i\in I}$ is a covering of $\mathcal{S}paces'_{fp,flat,proper}$ and every covering of $\mathcal{S}paces'_{fp,flat,proper}$ is isomorphic to one of these. Set $S_{ij} = S_i \times_S S_j$ and $X_{ij} = S_{ij} \times_S X$ so that $(X_{ij} \to S_{ij}) = (X_i \to S_i) \times_{(X\to S)} (X_j \to S_j)$. Next, suppose that $\mathcal{L}, \mathcal{N}$ are ample invertible sheaves on X/S so that $(X \to S, \mathcal{L})$ and $(X \to S, \mathcal{N})$ are two objects of $\mathcal{P}olarized$ over the object $(X \to S)$. To check descent for morphisms, we assume we have morphisms $(\mathrm{id}, \mathrm{id}, \varphi_i)$ from $(X_i \to S_i, \mathcal{L}|_{X_i})$ to $(X_i \to S_i, \mathcal{N}|_{X_i})$ whose base changes to morphisms from $(X_{ij} \to S_{ij}, \mathcal{L}|_{X_{ij}})$ to $(X_{ij} \to S_{ij}, \mathcal{N}|_{X_{ij}})$ agree. Then $\varphi_i : \mathcal{L}|_{X_i} \to \mathcal{N}|_{X_i}$ are isomorphisms of invertible modules over X_i such that φ_i and φ_j restrict to the same isomorphisms over X_{ij}. By descent for quasi-coherent sheaves (Descent on Spaces, Proposition 04W8) we obtain a unique isomorphism $\varphi : \mathcal{L} \to \mathcal{N}$ whose restriction to X_i recovers φ_i.

Decent for objects is proved in exactly the same manner. Namely, suppose that $\{(X_i \to S_i) \to (X \to S)\}_{i\in I}$ is a covering of $\mathcal{S}paces'_{fp,flat,proper}$ as above. Suppose we have objects $(X_i \to S_i, \mathcal{L}_i)$ of $\mathcal{P}olarized$ lying over $(X_i \to S_i)$ and a descent datum

$$(\mathrm{id}, \mathrm{id}, \varphi_{ij}) : (X_{ij} \to S_{ij}, \mathcal{L}_i|_{X_{ij}}) \to (X_{ij} \to S_{ij}, \mathcal{L}_j|_{X_{ij}})$$

satisfying the obvious cocycle condition over $(X_{ijk} \to S_{ijk})$ for every triple of indices. Then by descent for quasi-coherent sheaves (Descent on Spaces, Proposition 04W8) we obtain a unique invertible $\mathcal{O}_X$-module $\mathcal{L}$ and isomorphisms $\mathcal{L}|_{X_i} \to \mathcal{L}_i$ recovering the descent datum φ_{ij}. To show that $(X \to S, \mathcal{L})$ is an object of $\mathcal{P}olarized$ we have to prove that $\mathcal{L}$ is ample. This follows from Descent on Spaces, Lemma 0D3C.

Since we already have seen that $\mathcal{S}paces'_{fp,flat,proper}$ is a stack in groupoids over Sch_{fppf} (Lemma 0D1G) it now follows formally that $\mathcal{P}olarized$ is a stack in groupoids over Sch_{fppf}. See Stacks, Lemma 09WX. □